

FUNKSI DAN ADIEM 2 PEN GRAFKA.

$$y = \frac{2x^3}{x^2 - 1}$$

1) DEFINISI EREHUA $\text{Dom} f = \mathbb{R} - \{\pm 1\}$
 $x^2 - 1 \neq 0 \quad x \neq \pm 1$

2) ARDATUEKIN EBAKITU PUNNAK.

0x ardato $\rightarrow y = 0 \rightarrow 0 = \frac{2x^3}{x^2 - 1} \rightarrow x = 0 \rightarrow P(0,0)$

0y ardato $\rightarrow x = 0 \rightarrow y = 0$

3) SITENLIA

$f(-x) = \frac{2(-x)^3}{(-x)^2 - 1} = \frac{-2x^3}{x^2 - 1} = -f(x)$ SIM. BAKOLIA

4) PERIODIKOTASUMA Ez do.

5) ASINTOTAK.

A. bethko $x = 1$ do $x = -1$

$x = 1 \quad \lim_{x \rightarrow 1} f(x) = \left(\frac{2}{0}\right)$

$x = -1 \quad \lim_{x \rightarrow -1} f(x) = \left(\frac{2}{0}\right)$

$\frac{2}{0.99} \quad \frac{2}{1.01}$ $\left\{ \begin{array}{l} \lim_{x \rightarrow 1^-} \frac{2x^3}{x^2 - 1} = \frac{2}{0^-} = -\infty \\ \lim_{x \rightarrow 1^+} \frac{2x^3}{x^2 - 1} = \frac{2}{0^+} = +\infty \end{array} \right\}$ $\left\{ \begin{array}{l} \lim_{x \rightarrow -1^-} \frac{2x^3}{x^2 - 1} = \frac{-2}{0^+} = -\infty \\ \lim_{x \rightarrow -1^+} \frac{2x^3}{x^2 - 1} = \frac{-2}{0^-} = +\infty \end{array} \right\}$

A. 2EIMARRA

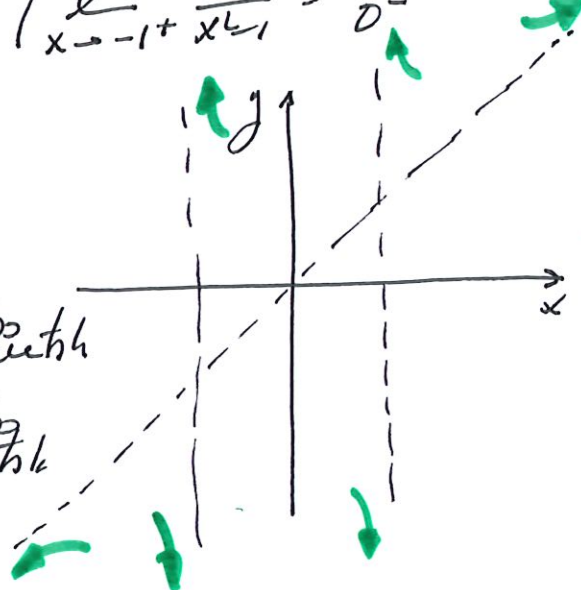
$\frac{2x^3}{x^2 - 1} = \frac{-2x^3 + 2x}{2x} \quad f(x) = 2x + \frac{2x}{x^2 - 1}$

$d = \frac{2x}{x^2 - 1}$ $\left\{ \begin{array}{l} x \rightarrow +\infty \quad d = \frac{+}{+} = 0^+ \text{ kurbo gelych} \\ x \rightarrow -\infty \quad d = \frac{-}{+} = 0^- \text{ kurbo at pihk} \end{array} \right.$

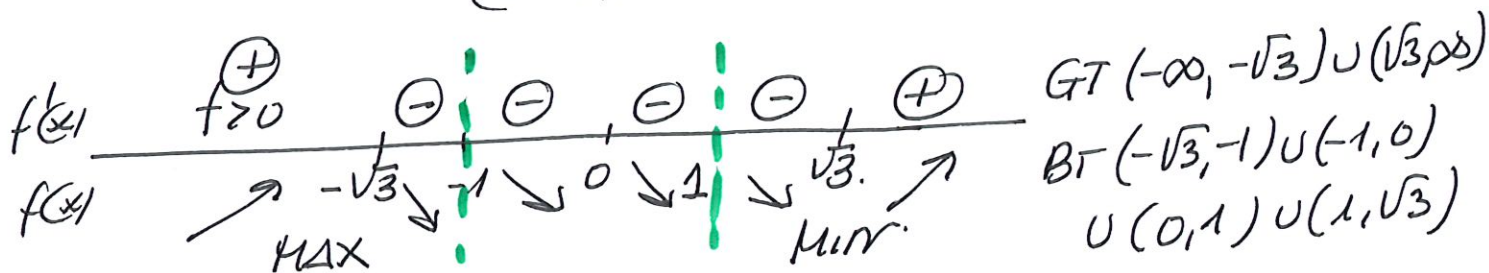
6) HAKKUNDA

$f(x) = \frac{2x^3}{x^2 - 1} \rightarrow f'(x) = \frac{6x^2(x^2 - 1) - 2x \cdot 2x^3}{(x^2 - 1)^2} = \frac{6x^4 - 6x^2 - 4x^4}{(x^2 - 1)^2}$

$f'(x) = \frac{2x^4 - 6x^2}{(x^2 - 1)^2} = \frac{2x^2(x^2 - 3)}{(x^2 - 1)^2}$



$$f'(x)=0 \rightarrow \frac{2x^2(x^2-3)}{(x^2-1)^2}=0 \rightarrow \begin{matrix} x=0 \\ x=\pm\sqrt{3} \end{matrix}$$



7.) GANBITASUNA / AMURTASUNA

$$f'(x) = \frac{2x^4 - 6x^2}{(x^2-1)^2}$$

$$\begin{aligned}
 f''(x) &= \frac{(8x^3 - 12x)(x^2-1)^2 - 2(x^2-1) \cdot 2x(2x^4 - 6x^2)}{(x^2-1)^4} = \\
 &= \frac{(x^2-1) [(8x^3 - 12x)(x^2-1) - 4x(2x^4 - 6x^2)]}{(x^2-1)^4} = \\
 &= \frac{\cancel{8x^5} - 8x^3 - 12x^3 + 12x - \cancel{8x^5} + 24x^3}{(x^2-1)^3} = \\
 &= \frac{4x^3 + 12x}{(x^2-1)^3} = \frac{4x(x^2+3)}{(x^2-1)^3}
 \end{aligned}$$

$$f''(x)=0 \rightarrow \frac{4x(x^2+3)}{(x^2-1)^3}=0 \rightarrow x=0$$

